

The Effects of Environmental Distress on Labor Markets: Evidence from Brazil

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Abstract. This article documents how environmental distress affects individual-level labor market outcomes in Latin America's largest economy. We collect data on a broad range of environmental distress events namely heat waves, floods, fires, and droughts, and combine these with uniquely rich administrative information covering the universe of formal employment in Brazil from 2003 to 2017. We find heterogeneous labor effects in response to environmental distress. We find that heat waves disrupt employment, increasing retirement rates by over 30 percent within the year of occurrence and beyond, driven by relatively more educated workers. Studying cross-municipality migration as a potential adaptation mechanism, we find little evidence that environmental distress increases the likelihood of individual out-migration. To the contrary, when floods or heat waves occur at typical destinations, the likelihood of migration among potential migrants and potential return migrants decreases significantly. Additionally, heat waves and floods reduce the likelihood cross-state out-migration, highlighting how environmental distress acts as a deterrent to migration. Our work contributes to understanding how environmental distress affects even the relatively more educated and higher-income formal labor force.

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1. Introduction

Climate change is increasing the frequency and intensity of natural disasters across the globe, with considerable economic consequences for affected populations (Aguilar-Gomez et al., 2024; Jay et al., 2023; Lee et al., 2023). Natural disasters have induced substantial labor reallocation and migration flows across regions (Albert et al., 2021; Baez et al., 2017; Mbaye and Zimmermann, 2015) and the effects are expected to increase as a consequence of global climate change. This study examines how environmental distress events like heat waves, floods, fires, and droughts affect individual-level labor market outcomes and migration decisions in Brazil by combining a uniquely rich administrative data set of employment links with geo-referenced information on environmental distress events. For the entire formal workforce, we observe job locations, start and end dates, monthly wages, and occupational codes of employment links for the years 2003-2017. This allows us to understand the causes of employment termination and detailed information on individual characteristics, which allows us to examine differential impacts across demographic groups.

We find that environmental distress events are disruptive to formal labor markets. Heat waves significantly increase the likelihood of employment termination within a given year. Evidence from time use data has shown that excessive heat reduces labor supply on the intensive margin (Graff Zivin and Neidell, 2014). Our findings point to additional reductions on the extensive margin: heat waves increase the likelihood of retirement by over 30 percent, not only in the year of the heat wave but also in subsequent years. This would be consistent with morbidity effects of heat (White, 2017). In line with previous work on the relationship between heat and mortality (Barreca et al., 2016; Deschênes and Greenstone, 2011), we find significant and persistent effects of heat waves on employment terminations caused by worker mortality. We do not find evidence that workers adapt by switching occupations, nor by migrating to other states or other municipalities, suggesting that workers affected by heat waves might experience labor frictions.

Migration is a prominent potential adaptation mechanism to unfavorable environmental conditions. Recent work by Borusyak et al. (2022) emphasizes that migration depends not only on conditions at the origin location, but also on conditions at potential destinations. We

incorporate these insights by simultaneously using information on the incidence of disasters at origin locations and at “typical” destination locations in our estimations. Including information from typical destinations reveals that heat waves and floods act as deterrents for potential migrants in multiple ways. First, they reduce the likelihood of cross-state migration among affected individuals. Second, when a wet-bulb heat wave or flood occur at a typical destination, the individual likelihood of out-migration decreases by 9.6 and 8.9 percent, respectively. Third, they reduce the likelihood of return migration to the affected municipality.

The increasing frequency of environmental disasters will likely disrupt economic activity in the future. This study contributes to our understanding of how environmental distress affects individual’s labor market outcomes and potential migration responses among the formal, relatively more educated and urbanized workforce. It is, to our knowledge, the first study to examine effects of environmental distress on individual-level responses at a national scale in a large country with a diverse range of climate zones, providing a compelling backdrop for studying impacts of environmental distress on labor markets.

Previous work on the effects of heat exposure shows compromise health among older-aged and high income earners (Ireland et al., 2023). Our results showing increases in retirement among older and higher income earners due to heat waves are consistent with these health effects. Evidence from time use data and survey production data has shown that excessive heat reduces labor supply on the intensive margin (Graff Zivin and Neidell, 2014; LoPalo, 2023). Heat exposure compromises individuals’ self-reported ability to work (Heyes and Saberian, 2022). However, the evidence on the impacts of other disasters such as floods and hurricanes is mixed. While strong hurricanes reduced workers’ earnings (Groen et al., 2020), and decreased growth rates and economic output (Strobl, 2011, 2012) some areas affected by hurricanes experience increased earnings as areas recover (Belasen and Polachek, 2009; Kocornik-Mina et al., 2020) or little to no changes in employment outcomes (Johar et al., 2022). Overall, the effects of natural disasters tend to be larger low-income countries (Felbermayr and Gröschl, 2014; Fomby et al., 2013; Noy, 2009). Therefore, studying the impacts of these disasters on labor outcomes in low- and middle-income countries is important.

This article also contributes to the literature that examines the impacts of climate and environmental distress on individual migration decisions. Studies in the U.S. have found that nat-

ural disasters induce out-migration from affected areas (Boustan et al., 2012, 2020; Deryugina et al., 2018). These effects are also documented in low- and middle-income countries (Albert et al., 2021; Cattaneo and Peri, 2016; Mueller et al., 2014; Quiñones et al., 2023). Our paper shows that adaptation to these labor shocks via migration might be limited: migration is affected by the characteristics of the potential destinations, restricting the relocation alternatives for affected workers. A closely related recent article is Albert et al. (2021), which shows out-migration from drought-affected regions in Brazil. We extend this work by studying other disasters and non-agricultural areas.

The remainder of this article is structured as follows: Section 2 details the data, section 3 lays out the estimation strategy, section 4 presents and discusses the results, and section 5 concludes.

2. Data

2.1. Environmental Stressors: Heat waves, Floods, Fires, and Droughts

We build a national database of environmental distress events from weather station data and remote sensing data. To obtain the occurrence of floods and heat waves, we use data from 558 weather stations operated by the National Meteorology Institute (INMET) in all 27 states. We focus our attention on municipalities that have weather stations, to avoid measurement error from data interpolation (Auffhammer et al., 2013). Figure 1 shows a map of Brazil municipalities, highlighting municipalities with available weather stations. In municipalities with weather stations, the average number of stations is 1.03. Throughout our study period, the meteorological institute INMET has continuously expanded their set of weather stations. We include municipalities in our sample only after their respective weather station has come online. The municipalities are shaded according to population size. The map illustrates that while we observe roughly 10 percent of municipalities, we observe a majority of highly populated municipalities, covering 41.3 percent of the population. We observe all 10 most populous municipalities, and 302 out of the top 1,000 most populous municipalities according to the 2010

Census. The average population size of a municipality in our sample is 144,979.¹

To define the occurrence of a flood, we combine information on precipitation and local flood susceptibility. A municipality experiences a flood when precipitation of 55mm or higher is detected within one hour², and the municipality has been classified as having a non-zero fraction of the population living in areas of high flood risk by the Brazilian Institute of Geographics and Statistics IBGE.³ We define that a heat wave occurs when the daily maximum temperature exceeds the local summer average by at least 6.4C for two or more consecutive days.⁴ Heat waves are more dangerous to human health in the presence of humidity, therefore we also calculate a humidity-adjusted temperature index,⁵ and define a "wet bulb heat wave" when the humidity-adjusted daily maximum temperature exceeds its historical summer average by at least 4.5C for two or more consecutive days.⁶

Information on wildfires, measured as count of fires per municipality per year, was obtained from the Global Fire Atlas data set published by NASA. The data set consists of shapefiles with information on daily fire dynamics from 2003 to 2017 of each ignition point. The data thus has information of the ignition, fire size, perimeter, speed, and direction of the spread. The underlying data includes the Moderate Resolution Imaging Spectroradiometer (MODIS) and burned area product (MCD64A1). We obtained the annual burned area at the municipality level by spatially merging Brazil municipalities and the Global Fire Atlas dataset and standardized the count of fires per municipality-year.

In order to classify droughts, we use the Standardized Precipitation and Evapotranspiration Index (SEPI), which is calculated from the difference between precipitation and potential evapotranspiration at different time scales. For the purpose of this paper, we obtained the 1-month SPEI index from the SPEI Global Drought Monitor. This dataset uses underlying data from the Climatic Research Unit of the University of East Anglia and provides a 0.5° gridded

¹The average population size of a municipality in Brazil is 34,278, with a standard deviation of 203,113 (Source: Decennial Census 2010).

²This precipitation threshold is used by the city of Rio de Janeiro to declare a state of emergency. Sistema Alerta Rio

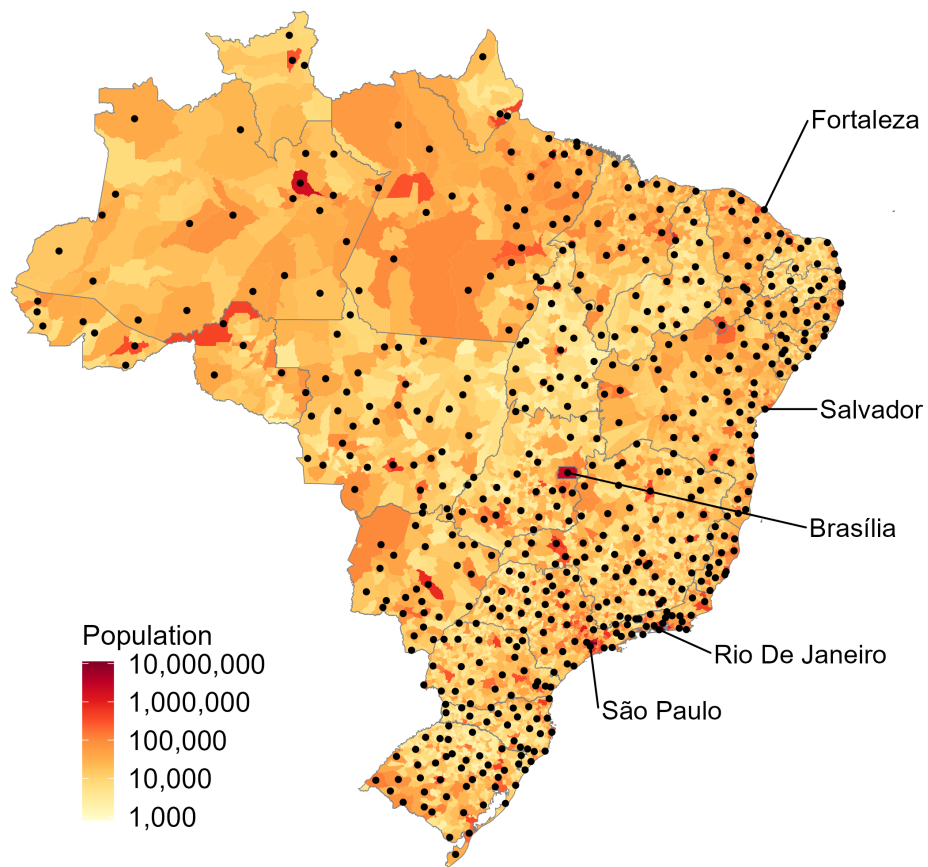
³IBGE Publication 101589 "Populacao em areas de risco no Brasil"

⁴We calculate the location-specific summer average temperature as follows: We average the daily maximum temperature per location per month, select the hottest month per location-year and average across years.

⁵<https://physicscalc.com/physics/wet-bulb-calculator/>

⁶To define heat waves, we follow the threshold values of 4.5C and 6.4C above local normal which is the definition used by the India Meteorological Department

Figure 1: Municipalities Observed in Sample



Notes: Municipalities with INMET weather stations are marked with dots. These form the study sample. Municipalities are shaded by population size according to the 2010 Decennial Census.

file with SPEI values for each pixel. We obtain the average monthly value at the municipality level. We classify moderate and extreme droughts using the U.S. Drought Monitor Classification⁷ and defined a year to have experienced a moderate or severe drought if it experienced a moderate or severe drought for 6 or more months.

2.2. Individual-level Location Histories

Our dataset contains the universe of formal employment in Brazil by year from 2003 to 2017. The primary unit of observation is a job with an individual identifier (“Cadastro de Pessoas Físicas”, CPF). We observe 95.6 million individuals in total. We randomly sample 10 million individuals for computational tractability.

The employment-level data offer rich detail: job start and end dates, average monthly pay, 6-digit occupational code, municipality, and demographic information. For terminated employment links, we observe the modality of termination, e.g. dismissal, resignation, retirement, or death of employee. We assign the municipality in January of year t as the initial location of job-holding individuals in year t . This is also the location which we use to measure whether an individual has been exposed to environmental distress in year t . If an individual holds multiple jobs within the same month, we assign the location of the highest-paying job. Within a given year, roughly 5 percent of individuals hold more than one job at the same time. We consider the likelihood of experiencing job termination, the modality of termination, and average monthly earnings as outcomes of interest. Additionally, we consider the likelihood of switching occupations and disappearance from the formal job market. ‘Disappearance’ can happen either if the individual remains unemployed, leaves the labor force, or moves to the informal sector.

To study migration, we define that a move occurred when the job municipality in Jan of year t is different from the job municipality in December of year t , conditional on having active employment links in both January and December of year t . Analogously, we define cross-state moves. Given the nature of our data, we consider this to be the best feasible approximation to true underlying location histories. We define return migration in year $t + 1$, conditional on

⁷<https://droughtmonitor.unl.edu/About/AbouttheData/DroughtClassification.aspx>

migration in year t , when the job municipality in December of $t + 1$ equals the one in January of year t .

We drop individuals younger than 16 or older than 70 years of age, observations with conflicting gender information, observations with missing educational information, and winsorize the top 1 percent observations of average monthly wages. We drop observations in municipalities without weather station information, and where we do not observe any emigrants in year $t - 1$. We use information on emigrants to construct a measure of disasters at potential destination locations, more detail is provided in Section 2.3.

Our final sample contains 5.6 million individuals residing in 538 municipalities in all 27 states. Table 1 presents summary statistics at the year-person level. The first column contains the summary statistics for the main estimation sample where we estimate the effects of disaster exposure on outcomes within the same year. In the second column we provide summary statistics for the year 2010, to be able to compare these to summary statistics from the Census 2010 in Table 2. Within a given year, 30.8% of individuals in the sample experience the termination of an employment link. 20.1% of employment links end in dismissal, 6.4% end due to quitting, 0.4% due to retirement, and 0.1% due to death of the employee. Another typical reason for employment termination is the end of the contractual period. Average monthly earnings total 2,801 in 2017 BRL, which is \$886 in 2017 USD.⁸ 15.6% experience a termination and are not observed in a new employment link by the end of the year, which we call “disappearance”. This can happen either if the person remains unemployed, exits the labor force, or works informally. Among all individuals who are observed in some active employment both in January and December of year t , we define occupational switches and migration. We define a switch in occupations when the job occupation code in January is different from the job occupation code in December. 8.9% of individuals in the sample switch occupations within a year.

We define migration as a difference in job locations between January and December of the same year. Given the nature of our data, we believe this is the best available approximation to actual migration. On average, 4.6% of individuals move across municipalities between January and December of a given year. 1.7% move across states. Among all individuals that were observed in January of year t and December of year $t + 1$, 0.7% had moved to a different mu-

⁸Average exchange rate of 2017: 3.1696

municipality but returned by December of year $t + 1$. From the Census data presented in Table 2, we learn that our migration measure overstates residential mobility. In the Census data, 1.8% of formally employed individuals who live in the same set of municipalities we observe in 2010 report living in their municipality of residence for less than one year. Our mobility measure captures switching jobs across municipalities, however, given the nature of our data, this is the only mobility measure that allows us to track millions of individuals year to year for over a decade.

Table 1 shows how frequently person-year observations experience heat waves, floods, fires, and droughts. On average, 9.5% of observations experience a heat wave, 12.7% experience a humidity-adjusted heat wave, 13.6% experience a flood. The typical fire exposure is -14.9% of a standard deviation. 13.7% of person-year observations experience a moderate drought, and 0.2% experience a extreme drought. In terms of the average population characteristics, the mean age in our sample is 36.9 years. 30.3% of person-year observations are younger than 30, 16.2% are older than 50 years. 30.3% have not completed high school, 49.0% have completed high school but not completed college, and 20.7% have completed college. 42.9% are female. We compare these characteristics in the dataset to the characteristics from 2010 Brazil Census. Table 2 shows the summary statistics for analogous samples constructed from the Brazilian 2010 Census. The first column shows the available summary statistics for the same set of municipalities we observe, considering only formally employed workers. The sample in column (1) is constructed to be as similar as possible to column (2) of Table 1. Employers are legally obliged to report details about employment links, but enforcement on the reporting of demographic characteristics of their employees is not very strict. This table shows that the RAIS sample is slightly older, with a smaller share of female workers and more educated workers. The average monthly earnings in constant 2017 BRL are 2,763 BRL in RAIS 2010, and 2,437 BRL in the Census 2010. At the 2017 exchange rate these amount to 872 and 769 USD. The average monthly earnings reported by employers to RAIS likely average over the 13th salary and vacation bonuses to which employees in Brazil are entitled, while the gross monthly wage income reported in the Census is elicited from individuals with reference to the month of July, and possibly subject to some under-reporting. The largest difference between the analogous samples is in the share of employees with a high school degree. It appears that 48.1% of observations in

RAIS 2010 have a high school degree, while only 39.6% have this degree in the analogous Census 2010 sample. While it is hard to imagine that individuals under-report their educational attainment in the Census, it is potentially more plausible that employees over-report their educational attainment to prospective employers. The second column shows the same summary statistics for both formally and informally employed workers, and the third and fourth column show the analogous characteristics for the total employed workforce in Brazil. We conclude that our study population has relatively high incomes, and is relatively more educated and, due to residing in relatively more populated municipalities, more urbanized than the general employed population.

In Appendix Table A1, we provide additional summary statistics for the samples used to estimate effects of disaster exposure on outcomes in years $t + 1$ and $t + 2$. These are smaller samples since we need to observe individuals in multiple periods.

2.3. Environmental Stressors at Potential Destinations

When individuals consider moving, the optimal choice depends not only on conditions at individuals' current locations, but also on conditions at potential destination locations. Recent work by Borusyak et al. (2022) emphasizes the importance of considering conditions at origins and destinations jointly when studying migration. To measure disasters at potential destinations, we construct a weighted average of disasters in other municipalities. We record all destinations of emigrants who left a given municipality in year $t - 1$, and calculate the shares of emigrant flows that each destination municipality received. As an example, suppose that 30 percent of emigrants who left municipality A in year $t - 1$ moved to municipality B , while all other emigrants moved someplace else. Suppose that only municipality B experiences a flood in year t . Then the value of "flood exposure at potential destination" for individuals in municipality A in year t is 0.3. Summary statistics on disaster exposure per person-year are presented in the lower panel of Table 1. The summary statistics suggest that migration destination locations are similar to origin locations with respect to disaster exposure. Typical destinations are slightly less likely to experience environmental distress.

Table 1: Summary Statistics

	Year t	2010
Number of year-person observations	28.2m	2.2m
Number of individuals	5.6m	2.2m
Number of municipalities	538	401
Number of states	27	26
Outcomes of Interest		
Share experience termination	0.308	0.317
Share dismissal	0.201	0.206
Share quit	0.064	0.069
Share retire	0.004	0.004
Share death	0.001	0.001
Mean monthly earnings in R\$2017	2,801.40	2,762.69
Share disappear	0.156	0.150
Share switch occupation	0.089	0.104
Share migrants	0.046	0.052
Share cross-state migrants	0.017	0.020
Demographics of Year-Person Observations		
Mean age in years	36.9	36.6
Share younger than 30	0.311	0.327
Share 30-50	0.527	0.519
Share 50 and older	0.162	0.154
Share less than high school degree	0.303	0.328
Share high school degree and some college	0.490	0.481
Share completed college	0.207	0.191
Share female	0.429	0.422
Disaster Exposure on Year-Person Level		
	Origin	Destination
Share heat wave	0.095	0.085
Share wetbulb heat wave	0.127	0.118
Share flood	0.136	0.115
Fires (SD)	-0.149	-0.146
Share moderate drought	0.137	0.136
Share extreme drought	0.002	0.002

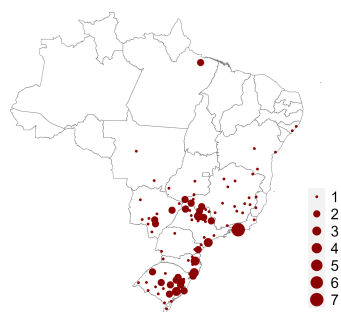
Notes: Descriptive statistics of the RAIS dataset for workers within the municipalities of our study.

Table 2: Summary Statistics Census 2010

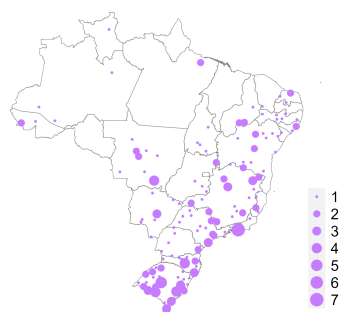
	Municipalities in sample, formal	Municipalities in sample, formal and informal	All munici- palities, formal	All munici- palities, formal and informal
Number of individuals	18.4m	24.2m	42.8m	58.8m
Number of municipalities	401	401	5565	5565
Number of states	26	26	27	27
Outcomes of Interest				
Mean monthly earnings in R\$2017	2,437.75 (3,204.01)	2,145.34 (2,963.82)	2,052.10 (2,568.04)	1,770.61 (2,346.75)
Share migrants	0.018	0.020	0.022	0.024
Share cross-state migrants	0.010	0.011	0.010	0.010
Demographics of Year-Person Observations				
Mean age in years	36.0 (11.5)	35.7 (11.8)	35.6 (11.5)	35.4 (11.8)
Share younger than 30	0.351	0.364	0.361	0.374
Share 30-50	0.501	0.485	0.499	0.482
Share 50 and older	0.149	0.151	0.140	0.143
Share less than high school degree	0.387	0.447	0.439	0.508
Share high school degree and some college	0.396	0.369	0.388	0.350
Share with college or more	0.217	0.184	0.173	0.141
Share female	0.453	0.470	0.433	0.449

Notes: The Census 2010 comparison sample consists of individuals who are actively employed, earning non-zero wages, aged between 16 and 70. No military, no sole proprietors, no employers.

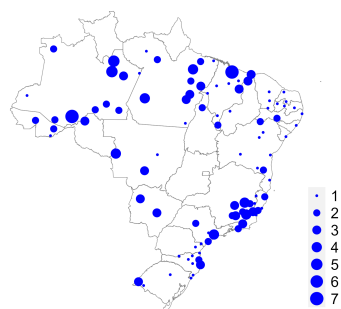
Figure 2: Spatial Distribution and Annual Frequency of Environmental Distress Events 2003-2017



(a) Heat waves



(b) Wetbulb heat waves



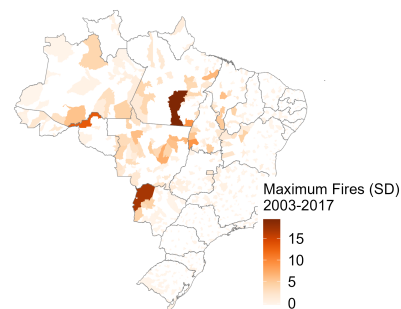
(c) Floods



(d) Moderate Droughts



(e) Extreme Droughts



(f) Fires

Notes: This figure shows the spatial distribution of environmental distress events across municipalities, summed across years in the study period 2003-2017.

3. Estimation

In order to examine the impact of environmental distress events on individual outcomes, we estimate the following equation:

$$y_{ijt} = \beta d_{ijt} + \mu_i + \nu_j + \gamma_t + \varepsilon_{ijt} \quad (1)$$

y_{ijt} is the outcome of interest of individual i in municipality j in year t . d_{ijt} is an indicator variable that equals one if a distress event occurred throughout year t in municipality j in which individual i was employed in January of year t . μ_i are individual fixed effects, ν_j are municipality fixed effects and γ_t are year fixed effects. ε_{ijt} denotes standard errors, clustered at the municipality level. Results of Equation 1 are presented in Table .

Recent work by Borusyak et al. (2022) formalizes how migration choices are driven by not only shocks at any given location, but simultaneously by shocks at potential destination locations. When shocks are spatially correlated, as is often the case for weather-related events, the coefficient estimate $\hat{\beta}$ will be biased. To address this concern, we estimate an individual-level version of Equation (22) in Borusyak et al. (2022). Taking the initial municipality j of individual i in year t as given, we construct a measure of exposure to disasters in potential destinations, $d_{it,-j}$, that individual i may consider by building a weighted average of disasters in all other locations. The weights are calculated as the share of emigrants from j who migrated to k in the previous year $t - 1$. We then estimate Equation 2. Results are presented in Tables 5, A4, and A5.

$$m_{ijt} = \beta_0 d_{ijt} + \beta_1 d_{it,-j} + \mu_i + \nu_j + \gamma_t + \varepsilon_{ijt} \quad (2)$$

m_{ijt} is an indicator that takes on the value of one if individual i is employed in December of year t in a municipality that is different from the municipality of employment in January of year t . We also investigate potential heterogeneity in impacts, whereby we repeat the analysis for different demographic groups: by age, by gender, and by educational attainment.

Table 3: Effects on environmental distress events on employment termination

	Termination	Dismissal	Quit	Retire	Death
Heat wave	0.0058** (0.0026)	0.0055 (0.0035)	0.0003 (0.0013)	0.0014** (0.0006)	0.0001 (0.0001)
Relative effect	1.9%	2.7%	0.5%	35.9%	8.6%
Wetbulb heat wave	0.0014 (0.0027)	-0.0002 (0.0036)	0.0013* (0.0007)	0.0002 (0.0006)	0.0001** (0.0000)
Relative effect	0.5%	-0.1%	2.0%	4.9%	9.2%
Flood	0.0054 (0.0040)	0.0052 (0.0034)	0.0005 (0.0006)	-0.0002 (0.0008)	0.0001*** (0.0000)
Relative effect	1.8%	2.6%	0.8%	-3.9%	7.0%
Fire	-0.0015 (0.0011)	-0.0015 (0.0013)	0.0003 (0.0002)	0.0001 (0.0001)	0.0000 (0.0000)
Relative effect	-0.5%	-0.7%	0.4%	2.9%	1.6%
Moderate drought	0.0026 (0.0021)	0.0018 (0.0032)	0.0008 (0.0008)	-0.0003 (0.0004)	-0.0001 (0.0001)
Relative effect	0.8%	0.9%	1.2%	-8.6%	-6.0%
Extreme drought	0.0031 (0.0080)	0.0006 (0.0074)	0.0010 (0.0030)	0.0023* (0.0013)	0.0001 (0.0001)
Relative effect	1.0%	0.3%	1.6%	57.2%	5.6%
Dependent variable mean	0.3076	0.2012	0.0637	0.0040	0.0011
N	28.2m	28.2m	28.2m	28.2m	28.2m
N individuals	5.6m	5.6m	5.6m	5.6m	5.6m

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The unit of observation is a person-year. Each coefficient estimate is based on a separate regression, with individual-level, municipality-level, and year-level fixed effects. We cluster standard errors at the level of municipalities.

4. Results

We find that heat waves significantly increase employment terminations. Heat waves increase the likelihood of experiencing a job termination by 1.9% over the baseline. Considering different modalities of termination, wet-bulb heat waves increase the likelihood of quitting. They also significantly increase the likelihood of death, as do flood events. We find that heat waves and extreme droughts increase the likelihood of retirement by 35.9 and 57.2%. Table 3 presents the results.

Looking at outcomes in the two years following an environmental distress event, we find

that heat waves continue to increase the likelihood of retirement not only within year t , but even in years $t + 1$ and $t + 2$ in similar magnitudes. We also find that heat waves and wet-bulb heat waves continue to increase the likelihood of death even into the next year relative to their occurrence. Tables A2 and A3 show the results.

Table 4 shows results for impacts on other employment-related outcomes. We find that flood events increase log earnings by 0.2866, which is in line with previous findings on wage increases after floods and hurricanes (Belasen and Polachek, 2009; Groen et al., 2020). Wet-bulb heat waves significantly increase the likelihood of an employment termination followed by no new formal employment by the end of the year, which we call "disappearance" from the formal labor market. We see no significant impacts on the likelihood of an individual switching occupations.

Consistent with existing literature, we show that droughts increase out-migration (Albert et al., 2021). Other environmental distress events at origin locations do not push individuals to out-migrate to other municipalities. If anything, we find that heat waves and floods reduce the likelihood to out-migrate to other states, by 4.9 and 4.2%, respectively. At the other end of the migration decision however, we do find that migration probabilities decrease significantly and considerably when distress events occur at typical destination locations. When floods occur at typical destination locations, the likelihood of individuals to out-migrate decreases by 8.9%. Wet-bulb heat waves at typical destinations reduce the likelihood of out-migration by 9.6%, droughts by 4.6%. We interpret this as evidence that environmental distress events render locations less attractive. Table 5 presents the results.

Effects on migration outcomes over time are presented in Tables A4 and A5. We find that heat waves at typical destinations inhibit return migration by $t + 1$ and in even larger magnitudes by $t + 2$. We do not find other effects on migration into the years $t + 1$ and $t + 2$, and interpret this as evidence that the inhibiting effects we find in year t are short-run effects, and migration picks up again afterwards.

Since we find considerable effects of heat waves on retirement, we explore further how the effects vary across demographic groups. Figure 4 shows the results. As expected, the effects are driven by relatively older workers. We do not find significant differences by gender. Considering different educational groups however, we see that the effect of heat waves on retirement is

Table 4: Effects of environmental distress events on other employment outcomes

	Log earnings	Disappear	Switch occup
Heat wave	-0.0588 (0.0654)	0.0024 (0.0021)	-0.0001 (0.0016)
Relative effect	-	1.5%	-0.1%
Wetbulb heat wave	0.0221 (0.0831)	0.0030* (0.0016)	-0.0002 (0.0005)
Relative effect	-	1.9%	-0.2%
Flood	0.2866* (0.1470)	0.0059 (0.0043)	0.0000 (0.0008)
Relative effect	-	3.8%	0.0%
Fire	-0.0434 (0.0472)	-0.0017 (0.0017)	-0.0004 (0.0004)
Relative effect	-	-1.1%	-0.5%
Moderate drought	-0.0616 (0.0672)	-0.0014 (0.0021)	0.0012 (0.0007)
Relative effect	-	-0.9%	1.3%
Extreme drought	-0.0652 (0.0725)	0.0011 (0.0043)	0.0009 (0.0031)
Relative effect	-	0.7%	1.0%
Dependent variable mean	7.5180	0.1556	0.0891
N	28.2m	28.2m	23.5m
N individuals	5.6m	5.6m	4.7m

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The unit of observation is a person-year. Each coefficient estimate is based on a separate regression, with individual-level, municipality-level, and year-level fixed effects. We cluster standard errors at the level of municipalities.

Table 5: Effects of environmental distress events on likelihood of migration

	Migration	Cross-state migration
Heat wave at origin	-0.0003 (0.0006)	-0.0008* (0.0004)
Relative effect	-0.5%	-4.9%
Heat wave at destination	-0.0018 (0.0015)	-0.0012 (0.0008)
Relative effect	-3.9%	-6.7%
Wetbulb heat wave at origin	0.0000 (0.0007)	-0.0003 (0.0006)
Relative effect	0.1%	-1.8%
Wetbulb heat wave at destination	-0.0044** (0.0022)	-0.0016* (0.0009)
Relative effect	-9.6%	-9.5%
Flood at origin	-0.0008 (0.0008)	-0.0007* (0.0004)
Relative effect	-1.6%	-4.2%
Flood at destination	-0.0041** (0.0018)	-0.0009 (0.0008)
Relative effect	-8.9%	-5.4%
Fire at origin	-0.0005 (0.0005)	-0.0003 (0.0003)
Relative effect	-1.1%	-1.9%
Fire at destination	0.0009 (0.0014)	0.0007 (0.0007)
Relative effect	2.0%	4.2%
Moderate drought at origin	0.0013*** (0.0004)	0.0003 (0.0003)
Relative effect	2.9%	1.9%
Moderate drought at destination	-0.0021* (0.0012)	0.0002 (0.0008)
Relative effect	-4.6%	1.0%
Extreme drought at origin	0.0028 (0.0038)	-0.0001 (0.0015)
Relative effect	6.2%	-0.9%
Extreme drought at destination	-0.0079 (0.0076)	0.0006 (0.0045)
Relative effect	-17.2%	3.4%
Dependent variable mean	0.0461	0.0172
N	23.9m	23.9m
N individuals	4.7m	4.7m

Notes: Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The unit of observation is a person-year. Each pair of origin-destination coefficient estimates is based on a separate regression as specified in Equation 2, with individual-level, municipality-level, and year-level fixed effects. We cluster standard errors at the level of municipalities.

driven by relatively more educated workers. Dividing our sample based on income quartiles, we find significant increases in retirement for the highest income earners. Taken together, our results suggest that heat waves generate attrition in the labor force by accelerating retirement especially among relatively more educated and high-earning workers.

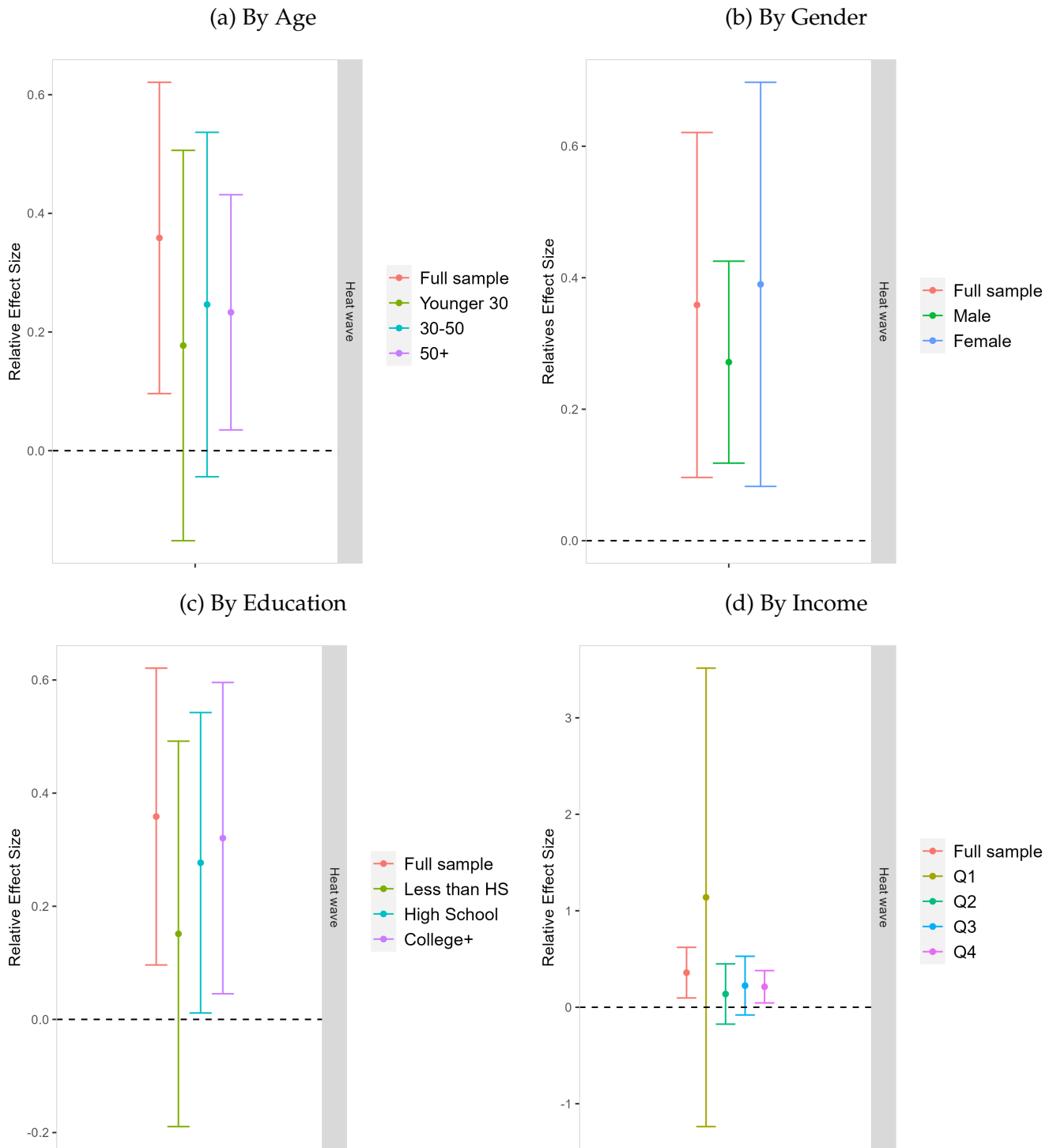
5. Conclusion

This study documents the effects of a broad range of environmental distress events on labor markets, focusing on granular individual-level outcomes. Given the nature of our data, a comprehensive annual individual panel covering all formally employed workers in Brazil, the focus is on the formal, urban workforce in a large developing country. We summarize our main findings as follows: heat waves disrupt employment links, and increase rates of retirement and mortality beyond the year of their occurrence. Studying different demographic groups, we find that the effect of heat waves on retirement are driven by relatively more educated workers, raising concerns about how climate change might reduce economic potential in aggregate terms.

We find little evidence of out-migration triggered by environmental distress events in the locations in which they occur. We do find that floods and heat waves act as deterrents to migration in multiple ways. In origin locations, they reduce the likelihood of cross-state migration. When they occur at typical destination locations, they reduce the likelihood of individual migration among potential immigrants, and among potential return migrants. The deterring effects appear to be short-lived, as they do not appear to deter migration by the end of subsequent years.

Our research highlights that environmental distress in the form of heat waves and floods disrupt employment links and deters potential immigration even among the formal, relatively educated and relatively urbanized workforce.

Figure 4: Effects of Heat Waves on Retirement Across Demographic Groups



Notes: Figure shows regression coefficients from specification 1 by age (Panel a), gender (Panel b), education (Panel c), and income (Panel d). Each coefficient estimate is based on a separate regression, with individual-level, municipality-level, and year-level fixed effects. We cluster standard errors at the level of municipalities.

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A. Appendix Tables

Table A1: Summary Statistics

	Year t	Year $t + 1$	Year $t + 2$	2010
Number of year-person observations	28.2m	23m	19.7m	2.2m
Number of individuals	5.6m	4.7m	4.3m	2.2m
Number of municipalities	538	498	471	401
Number of states	27	27	27	26
Outcomes of Interest				
Share experience termination	0.308	0.274	0.266	0.317
Share dismissal	0.201	0.173	0.165	0.206
Share quit	0.064	0.056	0.056	0.069
Share retire	0.004	0.005	0.005	0.004
Share death	0.001	0.001	0.001	0.001
Mean monthly earnings in R\$2017	2,801.40	3,068.92	3,218.46	2,762.69
Share disappear	0.156	0.130	0.123	0.150
Share switch occupation	0.089	0.265	0.375	0.104
Share migrants	0.046	0.099	0.131	0.052
Share return migrants	-	0.007	0.021	-
Share cross-state migrants	0.017	0.035	0.047	0.020
Cross-state return migrants	-	0.003	0.008	-
Demographics of Year-Person Observations				
Mean age in years	36.9	36.9	36.7	36.6
Share younger than 30	0.311	0.307	0.312	0.327
Share 30-50	0.527	0.535	0.538	0.519
Share 50 and older	0.162	0.158	0.150	0.154
Share less than high school degree	0.303	0.300	0.306	0.328
Share high school degree and some college	0.490	0.486	0.482	0.481
Share completed college	0.207	0.213	0.212	0.191
Share female	0.429	0.425	0.421	0.422
Disaster Exposure on Year-Person Level				
	Origin	Destination		
Share heat wave	0.095	0.085		
Share wetbulb heat wave	0.127	0.118		
Share flood	0.136	0.115		
Fires (SD)	-0.149	-0.146		
Share moderate drought	0.137	0.136		
Share extreme drought	0.002	0.002		

Notes: Descriptive statistics of the RAIS dataset for workers within the municipalities of our study.

Table A2: Effects of environmental distress events on employment termination in year $t + 1$

	Termination	Dismissal	Quit	Retire	Death
Heat wave	-0.0001 (0.0017)	0.0030** (0.0013)	-0.0021* (0.0011)	0.0018*** (0.0005)	0.0001** (0.0001)
Relative effect	0.0%	1.7%	-3.7%	37.1%	12.4%
Wetbulb heat wave	0.0007 (0.0015)	0.0024*** (0.0008)	0.0004 (0.0006)	0.0012*** (0.0003)	0.0001* (0.0000)
Relative effect	0.2%	1.4%	0.7%	24.9%	7.6%
Flood	0.0004 (0.0024)	0.0017 (0.0015)	0.0001 (0.0008)	-0.0004 (0.0008)	0.0001 (0.0001)
Relative effect	0.1%	1.0%	0.2%	-7.6%	7.5%
Fire	0.0008 (0.0005)	0.0006 (0.0004)	-0.0002 (0.0004)	0.0000 (0.0002)	0.0000 (0.0000)
Relative effect	0.3%	0.3%	-0.4%	1.1%	-1.5%
Moderate drought	0.0029 (0.0036)	0.0044 (0.0054)	0.0017* (0.0009)	-0.0006 (0.0008)	0.0000 (0.0001)
Relative effect	1.0%	2.6%	3.0%	-13.9%	2.2%
Extreme drought	0.0158*** (0.0050)	0.0029 (0.0032)	0.0059* (0.0034)	0.0027 (0.0021)	-0.0001 (0.0002)
Relative effect	5.8%	1.6%	10.4%	57.9%	-10.2%
Dependent variable mean	0.2738	0.1728	0.0562	0.0047	0.0012
N	23.0m	23.0m	23.0m	23.0m	23.0m
N individuals	4.7m	4.7m	4.7m	4.7m	4.7m

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The unit of observation is a person-year. Each pair of origin-destination coefficient estimates is based on a separate regression as specified in Equation 2, with individual-level, municipality-level, and year-level fixed effects. We cluster standard errors at the level of municipalities.

Table A3: Effects of environmental distress events on employment termination in year $t + 2$

	Termination	Dismissal	Quit	Retire	Death
Heat wave	0.0049** (0.0022)	0.0028 (0.0017)	-0.0020** (0.0008)	0.0017* (0.0010)	0.0000 (0.0001)
Relative effect	1.8%	1.7%	-3.5%	32.9%	2.0%
Wetbulb heat wave	0.0043** (0.0017)	0.0041*** (0.0012)	-0.0014*** (0.0005)	0.0014** (0.0007)	0.0001 (0.0001)
Relative effect	1.6%	2.5%	-2.5%	26.4%	4.5%
Flood	-0.0010 (0.0040)	-0.0017 (0.0032)	-0.0012* (0.0006)	0.0001 (0.0012)	0.0001 (0.0001)
Relative effect	-0.4%	-1.0%	-2.1%	2.5%	4.7%
Fire	0.0006 (0.0005)	0.0000 (0.0005)	0.0004* (0.0003)	0.0004 (0.0003)	0.0000 (0.0000)
Relative effect	0.2%	0.0%	0.8%	7.1%	-0.5%
Moderate drought	-0.0009 (0.0024)	0.0024 (0.0028)	-0.0002 (0.0006)	-0.0002 (0.0007)	0.0001 (0.0002)
Relative effect	-0.4%	1.5%	-0.4%	-3.8%	11.9%
Extreme drought	-0.0051 (0.0054)	-0.0043 (0.0030)	-0.0012 (0.0022)	0.0029 (0.0018)	-0.0004 (0.0003)
Relative effect	-1.9%	-2.6%	-2.2%	55.1%	-31.6%
Dependent variable mean	0.2662	0.1645	0.0558	0.0053	0.0013
N	19.7m	19.7m	19.7m	19.7m	19.7m
N individuals	4.3m	4.3m	4.3m	4.3m	4.3m

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The unit of observation is a person-year. Each pair of origin-destination coefficient estimates is based on a separate regression as specified in Equation 2, with individual-level, municipality-level, and year-level fixed effects. We cluster standard errors at the level of municipalities.

Table A4: Effects of environmental distress events on likelihood of migration by $t + 1$

	Migration by $t + 1$	Return migration	Cross-state migration by $t + 1$	Return cross-state migration
Heat wave at origin	0.0009 (0.0019)	-0.0001 (0.0002)	-0.0001 (0.0013)	-0.0001 (0.0001)
Relative effect	0.9%	-0.9%	-0.3%	-2.1%
Heat wave at destination	-0.0031 (0.0049)	-0.0007* (0.0004)	0.0005 (0.0011)	-0.0001 (0.0003)
Relative effect	-3.1%	-10.1%	1.4%	-4.1%
Wetbulb heat wave at origin	0.0009 (0.0015)	0.0000 (0.0001)	0.0001 (0.0016)	0.0000 (0.0001)
Relative effect	0.9%	0.4%	0.4%	-1.2%
Wetbulb heat wave at destination	-0.0050 (0.0050)	-0.0003 (0.0004)	-0.0022* (0.0013)	0.0003 (0.0003)
Relative effect	-5.1%	-4.4%	-6.4%	11.6%
Flood at origin	-0.0015 (0.0018)	0.0001 (0.0001)	-0.0009 (0.0009)	0.0000 (0.0001)
Relative effect	-1.5%	1.8%	-2.7%	-1.4%
Flood at destination	-0.0077 (0.0049)	-0.0003 (0.0004)	-0.0004 (0.0010)	-0.0002 (0.0003)
Relative effect	-7.8%	-4.0%	-1.1%	-6.5%
Fire at origin	-0.0006 (0.0011)	0.0000 (0.0001)	-0.0005 (0.0006)	0.0000 (0.0000)
Relative effect	-0.6%	0.6%	-1.4%	-1.0%
Fire at destination	0.0031 (0.0028)	0.0002 (0.0003)	0.0022* (0.0013)	0.0002 (0.0001)
Relative effect	3.1%	2.8%	6.3%	6.4%
Moderate drought at origin	0.0009 (0.0018)	-0.0004 (0.0002)	-0.0011 (0.0012)	-0.0001* (0.0001)
Relative effect	0.9%	-5.4%	-3.2%	-4.7%
Moderate drought at destination	-0.0026 (0.0038)	0.0010*** (0.0004)	0.0017 (0.0024)	-0.0003 (0.0002)
Relative effect	-2.6%	13.9%	4.8%	-9.7%
Extreme drought at origin	0.0119 (0.0101)	0.0001 (0.0008)	0.0006 (0.0037)	-0.0003 (0.0003)
Relative effect	12.0%	1.9%	1.8%	-9.7%
Extreme drought at destination	-0.0170 (0.0278)	-0.0028 (0.0033)	0.0092 (0.0118)	0.0000 (0.0009)
Relative effect	-17.2%	-40.3%	26.4%	0.0%
Dependent variable mean	0.0988	0.0070	0.0350	0.0028
N	20.0m	17.6m	20.0m	18.2m
N individuals	4.2m	3.7m	4.2m	3.8m

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The unit of observation is a person-year. Each pair of origin-destination coefficient estimates is based on a separate regression, with individual-level, municipality-level, and year-level fixed effects. We cluster standard errors at the level of municipalities.

Table A5: Effects of environmental distress events on likelihood of migration by $t + 2$

	Migration by $t + 2$	Return migration	Cross-state migration by $t + 2$	Return cross-state migration
Heat wave at origin	0.0009 (0.0031)	-0.0009 (0.0007)	-0.0006 (0.0012)	-0.0002 (0.0003)
Relative effect	0.7%	-4.4%	-1.2%	-3.0%
Heat wave at destination	0.0022 (0.0067)	-0.0049*** (0.0011)	-0.0024 (0.0019)	0.0000 (0.0006)
Relative effect	1.7%	-23.6%	-5.2%	-0.5%
Wetbulb heat wave at origin	0.0010 (0.0019)	-0.0007 (0.0009)	-0.0009 (0.0009)	-0.0003 (0.0003)
Relative effect	0.8%	-3.3%	-1.9%	-4.0%
Wetbulb heat wave at destination	0.0061 (0.0069)	-0.0049*** (0.0010)	-0.0004 (0.0034)	0.0001 (0.0004)
Relative effect	4.7%	-23.7%	-0.9%	1.1%
Flood at origin	0.0004 (0.0043)	-0.0004 (0.0006)	0.0006 (0.0017)	-0.0004** (0.0002)
Relative effect	0.3%	-1.7%	1.3%	-5.5%
Flood at destination	-0.0064 (0.0061)	-0.0018** (0.0009)	-0.0024 (0.0022)	0.0000 (0.0004)
Relative effect	-4.9%	-8.9%	-5.2%	-0.3%
Fire at origin	-0.0006 (0.0010)	0.0003** (0.0001)	-0.0006 (0.0006)	0.0003 (0.0003)
Relative effect	-0.4%	1.6%	-1.2%	3.4%
Fire at destination	0.0051 (0.0034)	-0.0003 (0.0013)	0.0033* (0.0019)	-0.0003 (0.0004)
Relative effect	3.9%	-1.5%	7.0%	-3.9%
Moderate drought at origin	0.0021 (0.0020)	0.0000 (0.0011)	0.0003 (0.0015)	0.0000 (0.0004)
Relative effect	1.6%	-0.1%	0.5%	0.0%
Moderate drought at destination	-0.0091 (0.0062)	0.0045 (0.0032)	0.0008 (0.0039)	-0.0010 (0.0010)
Relative effect	-6.9%	21.7%	1.6%	-13.4%
Extreme drought at origin	0.0179 (0.0175)	-0.0016 (0.0036)	0.0015 (0.0065)	-0.0002 (0.0012)
Relative effect	13.7%	-7.6%	3.2%	-2.2%
Extreme drought at destination	-0.0113 (0.0426)	-0.0002 (0.0090)	0.0080 (0.0136)	0.0025 (0.0036)
Relative effect	-8.6%	-1.2%	17.1%	32.3%
Dependent variable mean	0.1312	0.0206	0.0466	0.0078
N	17.3m	15.3m	17.3m	15.6m
N individuals	3.9m	3.5m	3.9m	3.5m

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The unit of observation is a person-year. Each pair of origin-destination coefficient estimates is based on a separate regression, with individual-level, municipality-level, and year-level fixed effects. We cluster standard errors at the level of municipalities.